

Freight Rate Prediction

ML Engineering Assessment | Spotter AI

August 2026

\$112.87

Ensemble MAE

5.20%

MAPE

12,000

Predictions

1. Data Exploration - Key Findings

The training dataset has 48,000 freight loads (Jan-Oct 2025) across three equipment types. Key findings:

- * distance has a 0.91 Pearson correlation with posted_rate -- the single dominant driver.
- * Average load: \$2,374 (median \$2,031, std \$1,486). Range: \$57 to \$25,533.
- * market_index and quote_signal are strong secondary predictors capturing live market conditions.
- * Equipment type matters: Reefer > Dry Van > Flatbed for typical per-mile rates.
- * Freight market peaked May-June 2025 (index ~1.30) then softened by August (~0.89).
- * Target (posted_rate) is right-skewed -- log-transform produces a near-normal distribution.

Dataset Overview

Training rows	48,000
Date range	2025-01-01 to 2025-10-31
Equipment types	Dry Van (56.7%) Reefer (25.1%) Flatbed (18.2%)
Mean posted_rate	\$2,374 (std \$1,486)
Median posted_rate	\$2,031
Max posted_rate	\$25,533
distance vs rate correlation	0.91 (strongest predictor)

2. Data Quality Issues & Resolutions

weight -- 300 missing (0.6%)

Filled with median per equipment type (not global median). Reefer and Dry Van trucks carry structurally different weights, so global imputation would introduce bias.

market_index -- 374 missing (0.8%)

Filled using a 50-row rolling mean on time-sorted data, then global median as fallback. Rolling mean respects temporal structure -- the market index on a given day is best estimated from nearby days.

Extreme posted_rate outliers -- 240 rows above \$6,511 (99.5th pct)

Capped at the 99.5th percentile for training only. These are real loads but allowing their full weight distorts gradient descent for the other 99.5% of data. Predictions are capped at 99.9th percentile (\$12,855) to prevent Ridge extrapolation from producing implausible rates.

3. Feature Engineering (33 features from 13 raw columns)

Time Features

- * day_of_week, month, quarter, day_of_year, week_of_year, is_weekend
- * days_since_ref: days since 2025-01-01 -- captures long-term market trend
- * Cyclical sin/cos of day-of-year, day-of-week, week-of-year -- ensures Dec 31 and Jan 1 are treated as adjacent

Market / Numeric Features

- * rate_per_mile = market_index / distance -- market pressure normalised by haul length
- * mi_x_qs = market_index x quote_signal -- interaction: when both are high, rates spike
- * dist_x_weight = distance x weight / 1M -- combined load effort (tonne-miles proxy)
- * log_distance, log_weight -- compress wide-ranging values for better model splits

Geographic Features

- * haversine_km -- true great-circle (crow-flies) distance between pickup and delivery
- * dist_ratio = road distance / haversine -- captures route tortuosity

Categorical Features

- * pickup, delivery, equipment -- native categories (LightGBM) or ordinal codes (Ridge/XGB)
- * route = pickup + delivery -- specific lane identifier (each lane has its own rate level)
- * weight_bin -- light / medium / heavy / very_heavy (structural pricing tiers)

Target: $\log(1 + \text{posted_rate_capped})$
Log-transform converts the skewed target to near-normal. At prediction time, `expm1()` reverses the transform.

4. Training & Validation Approach

Method: 5-Fold Rolling Time-Series Cross-Validation

A random split would allow training on October data and validating on January -- leaking future information. Instead, each fold trains on all past months and validates on the next unseen month:

Fold	Training Data	Validation Month
1	January - May 2025	June 2025
2	January - June 2025	July 2025
3	January - July 2025	August 2025
4	January - August 2025	September 2025
5	January - September 2025	October 2025

5. Model Selection & Comparison

Three model families were compared across all 5 folds:

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean MAE
Ridge Regression	\$156	\$111	\$100	\$140	\$117	\$124.68

LightGBM	\$179	\$123	\$127	\$138	\$127	\$138.94
XGBoost	\$202	\$131	\$137	\$135	\$136	\$148.34

Surprising finding: Ridge Regression outperformed both gradient boosting models.

After log-transforming the target, the pricing relationship becomes nearly linear:

$$\log(\text{rate}) \sim \log(\text{distance}) + \text{market_index} + \text{equipment_type} + \dots$$

Ridge's L2 regularisation prevents overfitting on the 10-month training window.

Final model: Weighted Ensemble (Ridge wins CV so gets highest weight)

$$\text{Final prediction} = 0.40 \times \text{Ridge} + 0.35 \times \text{LightGBM} + 0.25 \times \text{XGBoost}$$

Combining all three models reduces variance and gives better predictions than any single model alone.

6. Final Validation Metrics (Hold-Out: October 2025)

MAE - Mean Absolute Error	\$112.87	Average dollar error per load
MAPE - Mean Abs % Error	5.20%	Percentage error on average
RMSE - Root Mean Sq Error	\$651.21	Penalises large errors more heavily

The model predicts freight rates within 5.2% on average. On a load averaging \$2,374, that is roughly \$123 error per prediction.

Top 10 Most Important Features (LightGBM)

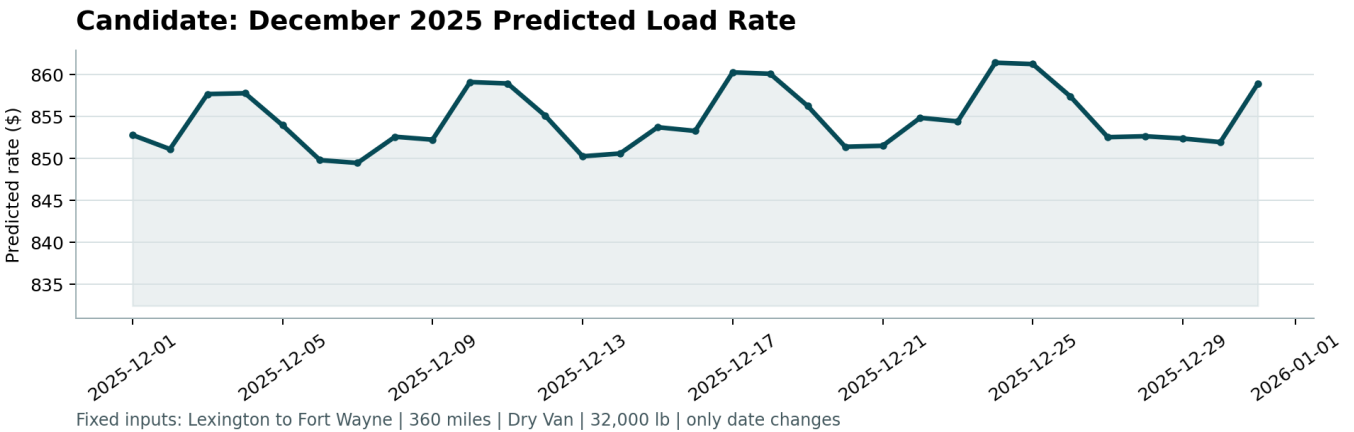
Rank	Feature	Why It Matters
1	distance	Primary cost driver -- longer haul = higher rate (r=0.91)
2	quote_signal	Live market quoting signal -- highly predictive of current rates
3	dist_x_weight	Combined load effort metric (distance x weight)
4	equipment	Reefer > Dry Van > Flatbed structural pricing difference
5	market_index	Overall market rate pressure index
6	pickup	Origin city lane-level rate effect
7	day_of_year	Seasonal effects -- Q4 is peak freight season
8	route	Specific origin-destination lane rates
9	delivery	Destination city lane-level rate effect
10	mi_x_qs	Interaction: high market_index AND high quote_signal

7. December 2025 Fixed-Route Predictions

Route: Lexington, KY -> Fort Wayne, IN | 360 miles | Dry Van | 32,000 lb

December 2025 is outside the training window so market_index and quote_signal are unknown. These were extrapolated using a linear trend fitted to Jan-Oct 2025 daily averages, plus a day-of-week offset to capture weekday vs weekend market patterns.

Predicted rate band: \$849 - \$861, with mild daily variation driven by weekday/weekend patterns.



8. Code & Submission Files

File	Description
train_and_predict.py	Full pipeline: clean, engineer, CV, ensemble, predict
validation_predictions.csv	12,000 final predictions (load_id, predicted_rate)
december-chart-inputs.csv	31-day December predictions (filled)
scorer_results/candidate_december.png	December predicted rate chart
score.py	Provided scorer and validator script
requirements.txt	Python package dependencies

GitHub: <https://github.com/Ca200905/freight-rate-prediction>